POWER BI DASHBOARD DOCUMENTATION

Smart Food Delivery Analytics

Dashboard screenshots • Step-by-step explanation • Issues and corrections

1. Project Overview

This report explains the Smart Food Delivery Analytics dashboard supplied as a PDF. The dashboard is intended to help review delivery distance, delivery time, traffic level, delay status, vehicle type, order type, and weather-related information. The screenshots below are included so that the explanation can be followed alongside the original dashboard.

2. Dashboard – Page 1



Figure 1. Original first-page dashboard screenshot supplied by the user.

Step-by-step explanation of Page 1

Step 1 – Dashboard title

"Smart Food Delivery Analytics" identifies the report topic. In the screenshot, the title is partly hidden/clipped at the top by the application notification/header area. Move the title down and check the report canvas before exporting.

Step 2 – Over All ID visual

This visual appears to use an ID field. An ID is normally a record identifier, so adding or plotting individual IDs does not clearly communicate the total number of orders. Replace it with a card using Count of Order ID or Distinct count of Order ID and title it "Total Orders".

Step 3 – Sum of Distance card

The card displays 142.12K. This is the sum currently shown by the report, but the value is meaningful only if summing the Distance field matches the intended business question. For average delivery distance, use Average of Distance and label the unit (for example, km) after verifying the source data.

Step 4 – Sum of Calculated_Distance_KM card

The card displays 1.13M. Verify that Calculated_Distance_KM is a numeric distance field and that its unit is kilometres. If the aim is typical delivery distance, use Average rather than Sum. Do not compare this total directly with the Distance total until their definitions and units are checked.

Step 5 – On Time & Delay Time cards

The screenshot shows Sum of Delay_Flag = 5K for Delayed and 0 for On Time. This may be caused by the flag encoding, filter context, or use of Sum for both categories. Check the Delay_Flag definition and compare with Delay_Status. Prefer separate order-count measures filtered by status.

Step 6 – Donut chart: Delay_Flag by Delay_Status and ID

The chart contains many tiny labels because ID is used as a category. This makes the visual cluttered and hard to interpret. Use Delay_Status as the category and show the count or percentage of orders in each status.

Step 7 – Delivery time by Traffic_Level

The chart compares Sum of Delivery_Time_Minutes across traffic levels such as High, Very High, Moderate, Low, and Very Low. A sum can be affected by how many orders are in each group. To compare delivery performance fairly, use Average Delivery Time by Traffic_Level and optionally add order count.

Step 8 – Delay flag by Distance_Band

The chart compares the summed delay flag for distance groups (Short, Medium, Long, Very Long). Confirm that Delay_Flag is coded consistently. Use delayed order count or delay rate for each band, with clear labels.

Step 9 – Delay flag by Delay_Status

This bar chart appears to show a large value for Delayed and zero for On Time. Confirm the flag values and filter settings. If the intention is to compare orders, plot order count by Delay_Status instead of summing a flag without validation.

Step 10 – Delivery_Time_Minutes visual

The lower-left visual shows a solid-looking bar and decimal/large x-axis values. The current field placement/aggregation does not clearly show a useful distribution. To show the distribution of delivery times, create delivery-time bins and plot order count by bin. To compare groups, use average delivery time by a category.

Step 11 – Delivery time by Delivery_Time and Order_Type

The visual includes a "#VALUE!" category and many decimal values. This suggests a data error or an unsuitable field/type arrangement. Inspect the source column, remove or correct invalid values, set the correct data type, and use meaningful time bands instead of many raw decimal categories.

Step 12 – Precipitation by Rating_Band

The chart shows percentage labels from 0% to 100% for rating bands. This may be a 100% stacked visual rather than a direct precipitation total. Decide whether the aim is to compare average precipitation or the percentage distribution of orders; then set the visual and title to match that goal.

3. Dashboard – Page 2

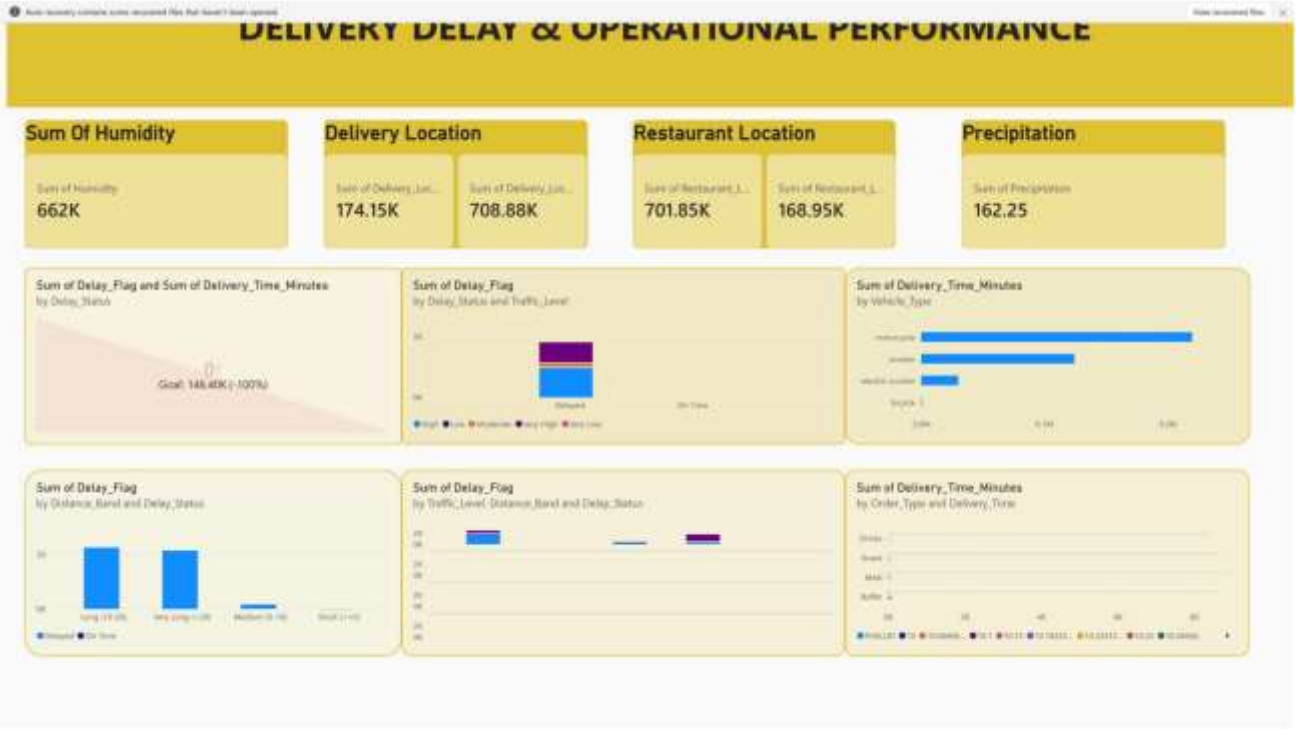


Figure 2. Original second-page dashboard screenshot supplied by the user.

Step-by-step explanation of Page 2

Step 13 – Humidity card

The card displays Sum of Humidity = 662K. Humidity is generally better summarized as an average or range, not a total. Confirm the dataset scale and unit, then consider "Average Humidity".

Step 14 – Delivery Location and Restaurant Location cards

The cards show large summed values for location-related fields. If these fields are latitude/longitude, their sums are not meaningful. Use a map and assign correct geographic data categories. If they are IDs or codes, use a count/distinct count or descriptive label. Confirm the actual column definitions first.

Step 15 – Precipitation card

The card displays Sum of Precipitation = 162.25. Confirm the unit and time period. If the objective is typical weather conditions, average precipitation may be more appropriate; only keep Sum if summing is valid for the dataset.

Step 16 – Delay_Flag and Delivery_Time_Minutes by Delay_Status

This visual compares a delay-related value and delivery time by status. Check whether both measures use suitable aggregations and have comparable units. Use order counts for volume and average minutes for delivery duration.

Step 17 – Delay_Flag by Delay_Status and Traffic_Level

This visual is intended to compare delay across traffic categories. Use delayed-order count or delay rate for each traffic level, rather than relying on a generic sum unless the flag coding is verified.

Step 18 – Delivery_Time_Minutes by Vehicle_Type

This chart compares delivery time across vehicle types. Use Average Delivery Time (minutes) by Vehicle_Type and optionally include order count so the audience can compare vehicle performance more clearly.

Step 19 – Delay_Flag by Distance_Band and Delay_Status

This visual compares delay information across distance bands and status. Use a clear count or percentage measure, and ensure the legend and title identify what is being measured.

Step 20 – Delay_Flag by Traffic_Level, Distance_Band and Delay_Status

This visual includes several categories at once and may be difficult to read. Keep only the most useful dimensions in one chart, or use small multiples/slicers to simplify comparisons.

Step 21 – Delivery_Time_Minutes by Order_Type and Delivery_Time

Check whether Delivery_Time is a category or a numeric value. If it is a duration, create meaningful bins; if it is a label, use it as a category. Avoid using a raw continuous field as a category when it produces many decimal labels.

Step 22 – Gauge/goal visual

The screenshot appears to show a value of 0 and a goal of 146.40K with -100%. Verify the value measure, goal measure, units, and filter context. Keep the gauge only if the goal is defined and meaningful; otherwise use a KPI card or variance visual.

4. Recommended Power BI correction workflow

1. Open the PBIX file in Power BI Desktop and select the relevant visual.

2. In the Visualizations / Build visual pane, inspect every field placed in Axis, Legend, Values, and Tooltips.

3. For identifier columns, choose Count or Distinct count instead of Sum, unless there is a documented reason to add the values.

4. For duration fields, use Average for typical delivery time or create bins for a distribution chart.

5. For Delay_Flag, confirm its coding (for example, whether 1 means delayed and 0 means on time) and compare it against Delay_Status.

6. In Power Query, inspect data types, blanks, duplicates, and error values such as #VALUE!. Correct them only according to the source-data meaning.

7. Rename visual titles in plain English and include units such as minutes or kilometres when confirmed.

8. Use the Filters pane and slicers to check whether filters are unintentionally excluding on-time orders.

9. Compare card totals with a table of the underlying order records before publishing or exporting.

10. Export to PDF and review that titles, labels, and charts are not clipped.

5. Suggested DAX measures (templates)

These are examples only. Replace Deliveries and Order_ID with the exact table/column names in the PBIX file. Confirm whether one row represents one order before using them.

Total Orders: `Total Orders = DISTINCTCOUNT(Deliveries[Order_ID])`

Delayed Orders: `Delayed Orders = CALCULATE([Total Orders], Deliveries[Delay_Status] = "Delayed")`

On-Time Orders: `On-Time Orders = CALCULATE([Total Orders], Deliveries[Delay_Status] = "On Time")`

Delay Rate: `Delay Rate = DIVIDE([Delayed Orders], [Total Orders], 0)`

Average Delivery Time: `Average Delivery Time (min) = AVERAGE(Deliveries[Delivery_Time_Minutes])`

Average Distance: `Average Distance = AVERAGE(Deliveries[Distance])`

Average Humidity: `Average Humidity = AVERAGE(Deliveries[Humidity])`

Average Precipitation: `Average Precipitation = AVERAGE(Deliveries[Precipitation])`

6. Suggested final dashboard layout

- KPI cards: Total Orders, Delayed Orders, On-Time Orders, Delay Rate, Average Delivery Time.
- Charts: Delay Rate by Traffic Level; Delay Rate by Distance Band; Average Delivery Time by Vehicle Type; Orders by Order Type.
- Weather analysis: average delivery time or delay rate by suitable weather bands, if those fields are valid.
- Location analysis: map only if location columns are confirmed as valid geographic coordinates.
- Slicers: date, traffic level, vehicle type, order type, and delay status, where those fields exist.
- Keep titles, units, colours, and number formats consistent. Remove unnecessary repeated visuals.

7. Data validation checklist

☐ Confirm the meaning and data type of each column.

☐ Check blank IDs, duplicate orders, missing values, and error values.

☐ Verify Delay_Flag against Delay_Status.

☐ Confirm units for distance, delivery time, humidity, and precipitation.

☐ Compare all headline cards with totals calculated from the source data.

☐ Test that slicers and cross-filtering update the charts consistently.

8. Conclusion

The supplied dashboard covers useful delivery-operation factors, but several visuals use Sum for fields that may be identifiers, flags, or environmental measurements. Other visuals include a #VALUE! category, many raw decimal categories, and unclear KPI aggregations. Validate the underlying data and column definitions, use suitable measures, and simplify the charts to make the report easier to understand.

9. Scope and limitation

This document is based on the two screenshots in the supplied PDF. The dataset and editable PBIX file were not provided. Therefore, the dashboard values and visual labels are described as they appear in the screenshots; the recommended corrections must be verified in Power BI against the real data model.